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Assignment #1

Q:No:1

$$y(n) - \frac{1}{3} y(n-1) + \frac{1}{5} y(n-2) = x(n) - 2x(n-1)$$

Sol:

By applying Z-Transform:

$$Y(z) - \frac{1}{3} z^{-1} Y(z) + \frac{1}{5} z^{-2} Y(z) = X(z) - \{1 - 2z^{-1})$$

$$Y(z)\left(1 - \tfrac{1}{3} z^{-1} + \tfrac{1}{5} z^{-2}\right) = X(z)\left(1 - 2z^{-1}\right)$$

$$\boxed{H(z) = \frac{Y(z)}{X(z)} = \frac{(1-2z^{-1})}{(1-\frac{1}{3}z^{-1}+\frac{1}{5}z^{-2})}}$$

We can write in Polynomial form as:

$$\boxed{H(z) = \frac{z^2 - 2z}{z^2 - \frac{1}{3}z + \frac{1}{5}}}$$

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Q: No: 2

$$x(n) = \delta(n) + \frac{1}{6}\delta(n-1) - \frac{1}{6}\delta(n-2)$$

$$y(n) = \delta(n) - \frac{2}{3}\delta(n-1)$$

Sol.

In z- domain.

$$X(z) = \underline{1 + 1/6\, z^{-1} - \frac{1}{6} z^{-2}}$$

$$Y(z) = 1 - \frac{2}{3} z^{-1}.$$

$$\boxed{H(z) = \frac{1 - 2/3\, z^{-1}}{1 + 1/6\, z^{-1} - 1/6\, z^{-2}}} \quad \text{--- (1)}$$

writing in Polynomial form.

$$\boxed{H(z) = \frac{2z(3z-2)}{6z^2 + z - 1}}$$

we can rewrite eq. (1) as: $\boxed{z^{-1} = x}$

$$\frac{1 - 2/3\, x}{1 + 1/6\, x - 1/6\, x^2} \quad \text{--- (2)}$$

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$$\frac{1-\frac{2}{3}x}{1+\frac{1}{6}x-\frac{1}{6}x^2} = \frac{A}{1-\frac{1}{3}x} + \frac{B}{1+\frac{1}{2}x}$$

$$1-\frac{2}{3}x = A\left(1+\tfrac{1}{2}x\right) + B\left(1-\tfrac{1}{3}x\right)$$

$$\boxed{B = 7/5} \qquad \because x = -2$$

$$\boxed{A = -2/5}$$

Put A & B in eq. 2

$$= \left(\frac{-2}{5}\right)\left(\frac{1}{1-\frac{1}{3}x}\right) + \left(\frac{7}{5}\right)\left(\frac{1}{1+\frac{1}{2}x}\right)$$

$$\boxed{H(z) = \frac{-2/5}{1-\frac{1}{3}z^{-1}} + \frac{7/5}{1+\frac{1}{2}z^{-1}}}$$

Impulse Response:

$$h(n) = \frac{-2}{5}\left(\frac{1}{3}\right)^n u(n) + \frac{7}{5}\left(\frac{-1}{2}\right)^n u(n)$$

Poles:

$$z = 1/3, \quad z = -1/2$$

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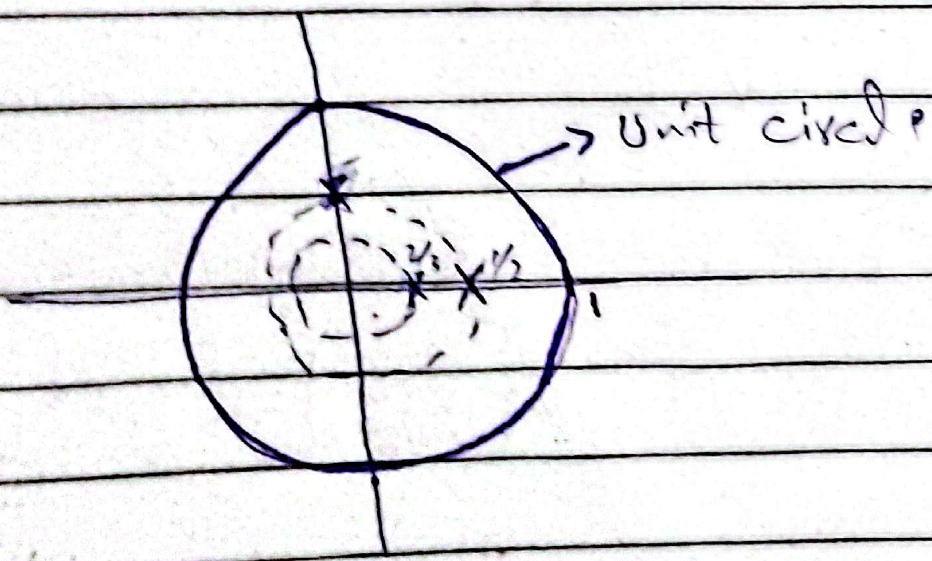
i) Frequency Response:

$$H(e^{j\omega}) = \frac{1-\frac{2}{3}e^{-j\omega}}{1+\frac{1}{6}e^{-j\omega}-\frac{1}{6}e^{-2j\omega}}$$

- stable (poles lies inside unit circle)

ii) System is stable



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Q: No: 3

$$H(z) = \frac{z+2}{2z^2+4-3z} \quad \text{(causal)}$$

$$H(z) = \frac{z^{-1}+2z^{-2}}{2-3z^{-1}+4z^{-2}}$$

Sol:

$$Y(z)(2-3z^{-1}+4z^{-2}) = X(z)(z^{-1}+2z^{-2})$$

Difference Equation:

$$y(n) = \frac{3}{2}y(n-1) - 2y(n-2) + \frac{1}{2}x(n-1) + x(n-2)$$

Frequency Response:

$$H(e^{jw}) = \frac{e^{jw}+2}{2e^{2jw}-3e^{jw}+4}$$

Stability:

$$b^2 - 4ac$$

$$(-3)^2 - 4(2)(4) = 9-3$$

$$= 6.$$

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Q: No: 4

$$x(n) = \left(\frac{1}{3}\right)^n u(n) - \frac{1}{5}\left(\frac{1}{3}\right)^{n-1} u(n-1)$$

$$y(n) = \left(\frac{1}{2}\right)^n u(n) \quad \text{(causal)}$$

Sol:

$$X(z) = \frac{1}{1-\frac{1}{3}z^{-1}} - \left(\frac{1}{5}z^{-1}\right)\left(\frac{1}{1-\frac{1}{3}z^{-1}}\right)$$

$$X(z) = \frac{1-\frac{1}{5}z^{-1}}{1-\frac{1}{3}z^{-1}}$$

$$Y(z) = \frac{1}{1-\frac{1}{2}z^{-1}}$$

Transfer function:

$$H(z) = \frac{1-\frac{1}{3}z^{-1}}{(1-\frac{1}{2}z^{-1})(1-\frac{1}{5}z^{-1})}$$

Impulse Response:

$$H(z) = H(x) = \frac{1-\frac{1}{3}z^{-1}}{(1-\frac{1}{2}z^{-1})(1-\frac{1}{5}z^{-1})}$$

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$$1-\frac{1}{3}x = A\left(1-\frac{1}{5}x\right)+B\left(1-\frac{1}{2}x\right)$$

$\boxed{B = 4/9}$ $\quad \because x = 5$

$\boxed{A = 5/9}$ $\quad \because x = 2$

$$H(x) = \frac{5}{9}\left(\frac{1}{1-\frac{1}{2}x}\right)+\frac{4}{9}\left(\frac{1}{1-\frac{1}{5}x}\right)$$

$$H(z) = \frac{5/9}{1-\frac{1}{2}z^{-1}} + \frac{4/9}{1-1/5\,z^{-1}}$$

Impulse Response:

$$h(n) = \frac{5}{9}\left(\frac{1}{2}\right)^n u(n) + \frac{4}{9}\left(\frac{1}{5}\right)^n u(n)$$

Stability:

Yes the system is stable.

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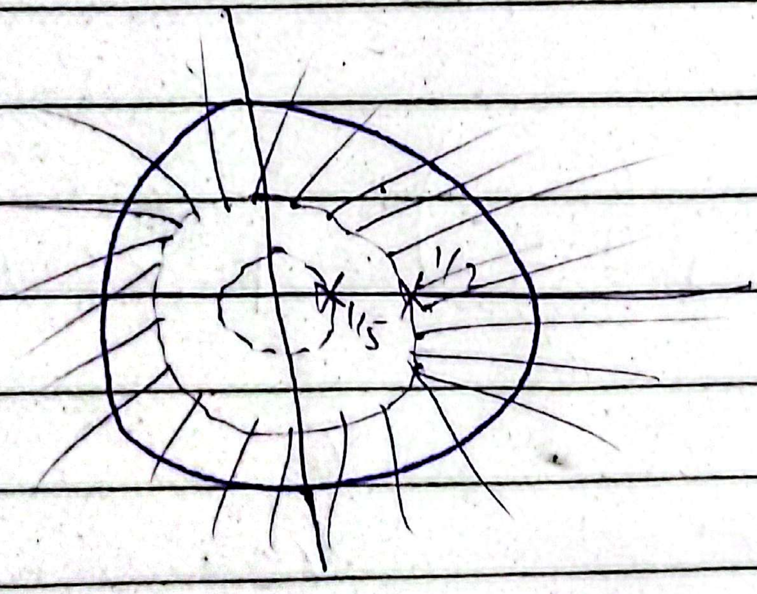
Frequency Response

$$H(e^{jw}) = \frac{1 - \frac{1}{3}e^{-jw}}{\left(1-\frac{1}{2}e^{-jw}\right)\left(1-\frac{1}{5}e^{-jw}\right)}$$



As unit circle is under ROC system is stable.